

Voxelization

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I. INTRODUCTION

Voxelization is the process of “discretizing” a 3D object into an axis-aligned grid. Besides being used stylistically in video games, this kind of representation is also used in many fields, such as medical imaging, urban analysis, computer graphics, and simulations. This report explores the design process of a parallel voxelization algorithm, which is implemented in C++ and utilizes the OpenMP library, taking advantage of multi-core processing to enhance performance.

II. DEVELOPMENT

A. Algorithm Description

The goal of the algorithm is to generate a representation of the 3D object as a set of voxels (volumetric pixels). A volumetric resolution is defined, which determines the number of voxels used along each axis.

The main idea is to iterate through the model’s triangles and paint the voxels they intersect with. There are several ways to determine this voxel set, and a quick-to-implement approach is to check all voxels inside a triangle’s bounding box.

B. Paradigm Choice

This voxelization algorithm executes the same “check-and-paint” procedure for each triangle, which makes it a good candidate for the task parallelization paradigm. The model’s triangles are distributed among the available processes, checked, and then used to paint the corresponding voxels. Each local result can then be merged to produce the voxelized output.

C. PRAM

The theoretical PRAM model is used to build a high-level version of the algorithm, which considers infinite processes. The algorithm is as follows:

Algorithm 1 Voxelization (PRAM)

```
1: triangles  $\leftarrow$  triangulate(model.load(path)).faces
2: for all triangle in triangles pardo do
3:   for all voxel inside bounding box of triangle do
4:     paint voxel
5:   end for
6: end for
7: rebuild 3D model using voxels
```

III. RESULTS AND ANALYSIS

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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